

**COURSE-0000 — Institution**

# **Course Title**

**Lecture Notes**

Your Name

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## Using this template

|                 |                                   |
|-----------------|-----------------------------------|
| <b>Lecture</b>  | Lesson 1 (Introduction)           |
| <b>Textbook</b> | H&P [1] §1.1–1.3                  |
| <b>Keywords</b> | sidenotes, index, bilingual build |

This chapter is a showcase of every macro the class provides. Delete it when you start a real course, but keep `02-lesson-skeleton.tex` as a starting point for new lessons.

### 1.1 Margin notes

The right-hand column is for anything that supports the main text without interrupting it. Numbered remarks use `\sidenote`<sup>1</sup> and unnumbered remarks use `\margin`.

Three labelled variants make the *kind* of side content visible at a glance:

- `\source{...}` – where the material comes from (lecture, textbook section, paper);
- `\tip{...}` – practical advice;
- `\caution{...}` – pitfalls and misconceptions.

### 1.2 Terms and the index

A *pipeline* is introduced with `\term`, which emphasizes the word, adds it to the index, and optionally prints a gloss in the margin (here: the Japanese term). Later mentions of a *pipeline* use `\term*` so the index does not fill up with repeated page numbers. Terms can be nested: *speedup* and *Amdahl's law* both go to the index.

**Definition 1.1 (Speedup).** For a change that makes a fraction  $f$  of the execution time  $s$  times faster, the overall speedup is

$$S = \frac{1}{(1-f) + f/s}. \quad (1.1)$$

**Example 1.2.** If  $f = 0.9$  and  $s \rightarrow \infty$ , Equation (1.1) gives  $S = 10$ : the untouched 10 % bounds the achievable speedup.

**Note 1.3.** definition, example and note share one counter per chapter, so they can be cross-referenced with `\cref`.

#### Key point – Amdahl's law

Optimize the common case; the part you do not touch limits the gain. Margin macros work inside this box too.

### 1.3 Figures, tables and code

Small figures go in the margin (Figure 1.1); normal figures use the standard figure environment; wide figures use `widefigure`, which spans the body and the margin column (Figure 1.3). Figures shared between the two editions live in `figures/` and switch their labels with `\iflangja`.

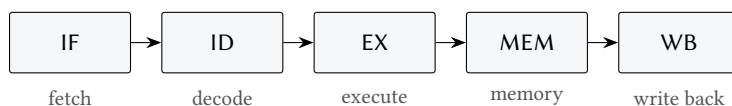


Figure 1.2: A classic five-stage pipeline (shared TikZ source).

<sup>1</sup> Like this one. Numbers restart at every chapter.

An unnumbered margin note.

**Source** Amdahl's original paper [2]; H&P §1.9.

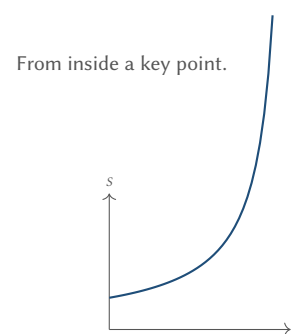
**Tip** Use `\tip` for exam-relevant advice or mnemonics.

**Caution** Use `\caution` for common misconceptions.

▷ パイプライン

▷ 高速化率

▷ Amdahl の法則



From inside a key point.

Figure 1.1: A figure in the margin (marginfigure).

| Cycle | 1  | 2  | 3  | 4   | 5   | 6   | 7  |
|-------|----|----|----|-----|-----|-----|----|
| add   | IF | ID | EX | MEM | WB  |     |    |
| sub   |    | IF | ID | EX  | MEM | WB  |    |
| lw    |    |    | IF | ID  | EX  | MEM | WB |

**Figure 1.3:** A wide float spanning body and margin (widefigure).

Code uses listings:

**Listing 1.1:** A loop in assembly.

```

1 | loop:   add    r1, r1, r2      ; accumulate
2 |         addi   r3, r3, -1
3 |         bne    r3, r0, loop    ; branch if not done

```

Pseudocode uses algorithmicx:

---

#### **Algorithm 1** Two-bit saturating counter update

---

```

1: if taken then
2:    $c \leftarrow \min(c + 1, 3)$ 
3: else
4:    $c \leftarrow \max(c - 1, 0)$ 
5: end if

```

---

## 1.4 Bilingual writing

Both editions share `format/`, `figures/` and `references.bib`; only the prose under `en/` and `ja/` differs. Inside shared files use `\iflangja{japanese}{english}` or the shorthands `\ja{...}` and `\en{...}`.

## Lesson skeleton

|                 |                         |
|-----------------|-------------------------|
| <b>Lecture</b>  | Lesson 2, videos 1–12   |
| <b>Textbook</b> | H&P [1] ch. 1           |
| <b>Keywords</b> | first term, second term |

Two to four sentences: what this chapter covers, why it matters, and how it continues the previous chapter.

### 2.1 First topic

The first sentence of the section. A *first term* is defined where the source defines it. The explanation follows the order of the source: what the concept is, why it is needed, and how it is used.

**Definition 2.1** (First term). A precise statement of the definition.

**Example 2.2.** A worked example with our own numbers.

### 2.2 Second topic

Body text that builds on Section 2.1. Each margin item sits next to the sentence it supports, and a short run of text carries at most about three of them.

A rule of thumb goes into a tip. The argument continues in the body, so that a reader who skips the margin still follows it. A longer aside that needs more than three lines goes into a numbered note.<sup>1</sup>

#### Key point – Lesson summary

- The one thing to remember from the first section.
- The one thing to remember from the second section.

**Source** Lesson 2, videos 1–5; H&P §1.2.

▷ 最初の用語

Notation and units used in the example.

**Source** Lesson 2, videos 6–12.

**Tip** A rule of thumb.

<sup>1</sup> Why the design exists, or a named real-world implementation, with its source.

## Bibliography

- [1] John L. Hennessy and David A. Patterson. *Computer Architecture: A Quantitative Approach*. 6th ed. Morgan Kaufmann, 2017.
- [2] Gene M. Amdahl. “Validity of the Single Processor Approach to Achieving Large Scale Computing Capabilities”. In: *AFIPS Conference Proceedings* 30 (1967), pp. 483–485.

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